ENG 101
Lesson 19

1 Main idea:

Which statements do not express the main idea of the text?
1. Core memory was the first type of computer memory developed.
2. There are at least three different kinds of memory used in computers.
3. Bubble memory is the latest development in computer memory.

2 Understanding the passage:

Decide whether the following statements are true or false (T/F) by referring to the information in the text. Then make the necessary changes so that the false statements become true.

T	F	
☐	☐	1. The most important function of a computer is to hold information in its memory in order to process it.
☐	☐	2. Minicomputers, microcomputers, and mainframes all have the same kind of memory.
☐	☐	3. Semiconductor memory was developed before core memory and after bubble memory.
☐	☐	4. Core memory uses small metal rings which can be magnetized or unmagnetized.
☐	☐	5. The state of the core can be represented by either 0 or 1.
☐	☐	6. Early computer memories had less storage capacity than newer ones.
☐	☐	7. A transistor and a chip are the same kind of device.
☐	☐	8. The development of chips make it possible for minicomputers and microcomputers to be invented.
☐	☐	9. Bubble memory is smaller than a chip.
☐	☐	10. Bubble memory doesn't have very many advantages.

3 Locating information

Find the passages in the text where the following ideas are expressed. Give the line references.

…………1. First there is core memory.
…………2. Further to this development, chips evolved.
…………3. There are three types of memory board.
…………4. This consists of producing a thin film over a memory board.
…………5. Then semiconductor memory was developed.
…………6. There is still a lot to learn about this process.
…………7. This is made up of thin wires and rings.
…………8. Finally, bubble memory was invented.

4 Contextual reference

Look back at the text and find out what the words in **bold** typeface refer to.

1. Long enough to process **it** (1.3)……….
2. **where** the wires crossed (1.7)……….
3. w**hich** was capable of being (1.9)……….
4. By **their** receptive addresses (1.12)……...
5. **This** has been made possible (1.6)………
6. **which** revolutionized the computer field (1.21)……….
7. **each one** capable of storing one bit (1.25)……….
8. of the circuits etched on **it** (1.26)……….
9. **it** produces magnetic bubbles (1.37)……...
10. **of which** represents one bit (1.38)……….

5 Understanding words

Refer back to the text and find synonyms for the following words.

1.said (1.1)…………...
2.own (1.2)…………...
3.progress (1.17)…………...
4.keeping (1.33)…………...
5.appropriate (1.42)…………...

Now refer back to the text and find antonyms for the following words.

6. neither … nor (1.1)…………...
7. bypassed (1.2)…………...
8. increased (1.17)…………...
9. Not producing (1.33)…………...
10. don't use up (1.42)…………...

6Words forms

First choose the appropriate form of the words to complete the sentences. Then check the differences of meaning in your dictionary.

1. alteration, alter, altered
a.When a program doesn't work properly, it is often necessary to make……..to it.
b.The omission of data from a program can…………its results drastically.
c. The use of the computer in business has………..the workload of many people.
2. electricity, electric, electrical, electrically
a.A lot of…………is needed to operate large computer systems.
b. Alexander Graham Bell invented the……………light bulb.
c.Many students today are studying to become……………engineers.
3. reduction, reduce, reduced
a.The introduction of the computer in the workplace has…………..the workload of many people.
b. There will probably be a great…………….in the consumption of oil in the next decade due to the use of computer technology.
4. creation, create, created, creative
a.A programmer usually has a…………as well as a logical mind.
b.It takes a lot of inspiration and hard work to come up with a new…………in computer technology.
c. Computers have certainly …………….few opportunities for fraud.

7Content review

Use the information in the text on 'Types of Memory' to complete the table.

Type	Developed	Size	Composition	Memory Capacity
1.		Large		80,000 bits

2. Integrated circuits on non-metallic element

3.

8Focus review

Focus G Time sequence

Complete the following table by referring back to the text on ‘Types of Memory’.

Para	Time Sequence Marker	Information
[2]	The computer	had memories up of a kind of grid of fine vertical and horizontal wires
[2]	computers	had a capacity of around 80,000 bits
[2]	whereas	it is not surprising to hear about computers with a memory capacity of millions of bits
[2]		core memory dominated the market
[3]		there was further development which revolutionized computer field

Focus I Adding information:

Complete the following sentences by referring back to the text on ‘Types of Memory’.

1. In the 1970s there was a …………………. development which revolutionized the computer field. (Para. 3)
2. …………………., the size of the components containing the circuitry can be considerably reduced… (Para. 3)
3. ………………….development in the field of computer memories is bubble memory. (Para. 4)

Exercise 1

Study the following definitions. A definition usually includes all three parts: the term to be defined, the group it belongs to, and the characteristics which distinguish it form other members of the group.

TERM	GROUP	CHARACTERISTICS
A core	is a ferrite ring	which is capable of being

		either magnetized or demagnetized
Silicon	is a nonmetallic element	with semiconductor characteristics

Now analyse the following definitions and identify the different parts by circling the term; by underlining the group once, and by underlining the characteristics twice.

1. A computer is a machine with an intricate network of electronic circuits that operate switches or magnetize tiny metal cores.
2. An abacus is a bead frame in which the beads are moved form left to right
3. Input is the information presented to the computer.
4. The term ‘computer’ includes those parts of hardware in which calculations and other data manipulations are performed, and he high-speed interval memory in which data and calculations are stored during actual executions of programs.
5. A ‘system’ is a good mixture of integrated parts working together to form a useful whole.
6. Large computer systems, or mainframes, as they are referred to in the field of computer science, are those computer systems fund in computer installations processing immense amounts of data.
7. Although there is no exact definition for a minicomputer, it is generally understood to refer to a to a computer whose mainframe is physically small, has a fixed word length between 8 and 32 bits, and costs less than $100,00 for the central processor.